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Studierichting: Informatica

Oefeningenreeks 4A nr. 48.1

$$\begin{aligned}\int \frac{\text{Bgtan } x}{x^2} dx &= - \int \text{Bgtan } x d\left(\frac{1}{x}\right) = -\frac{\text{Bgtan } x}{x} + \int \frac{1}{x} d(\text{Bgtan } x) \\&= -\frac{\text{Bgtan } x}{x} + \int \frac{1}{x(x^2 + 1)} dx \\&\rightarrow \frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} \\A &= \frac{1}{1} = 1\end{aligned}$$

$$\begin{aligned}B(i) + C &= \frac{1}{i} = -i \Rightarrow B = -1, C = 0 \\&= -\frac{\text{Bgtan } x}{x} + \int \frac{1}{x} dx + \int \frac{-x}{1 + x^2} dx \\&= -\frac{\text{Bgtan } x}{x} + \ln|x| - \frac{1}{2} \int \frac{1}{1 + x^2} d(x^2) \\&= -\frac{\text{Bgtan } x}{x} + \ln|x| - \frac{1}{2} \ln|1 + x^2| + c\end{aligned}$$

References